Which optimization algorithm to choose?

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2022-03-10

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1 Quick overview of main optimization methods

We present very quickly the main optimization methods. Please refer to **Numerical Optimization (Nocedal & Wright, 2006)** or **Numerical Optimization: theoretical and practical aspects (Bonnans, Gilbert, Lemarechal & Sagastizabal, 2006)** for a good introduction. We consider the following problem $\min_x f(x)$ for $x \in \mathbb{R}^n$.

1.1 Derivative-free optimization methods

The Nelder-Mead method is one of the most well known derivative-free methods that use only values of f to search for the minimum. It consists in building a simplex of $n+1$ points and moving/shrinking this simplex into the good direction.

1. set initial points $x_1, \dots, x_{n+1}$.
2. order points such that $f(x_1) \leq f(x_2) \leq \dots \leq f(x_{n+1})$.
3. compute x_o as the centroid of $x_1, \dots, x_n$.
4. Reflection:
   - compute the reflected point $x_r = x_o + \alpha(x_o - x_{n+1})$.
   - **if** $f(x_1) \leq f(x_r) < f(x_n)$, then replace x_{n+1} by x_r, go to step 2.
   - **else** go step 5.
5. Expansion:
   - **if** $f(x_r) < f(x_1)$, then compute the expansion point $x_e = x_o + \gamma(x_o - x_{n+1})$.
   - **if** $f(x_e) < f(x_r)$, then replace x_{n+1} by x_e, go to step 2.
   - **else** x_{n+1} by x_r, go to step 2.
   - **else** go to step 6.
6. Contraction:
   - compute the contracted point $x_c = x_o + \beta(x_o - x_{n+1})$.
   - **if** $f(x_c) < f(x_{n+1})$, then replace x_{n+1} by x_c, go to step 2.
   - **else** go step 7.
7. Reduction:
   - for $i = 2, \dots, n+1$, compute $x_i = x_1 + \sigma(x_i - x_1)$.

The Nelder-Mead method is available in `optim`. By default, in `optim`, $\alpha = 1, \beta = 1/2, \gamma = 2$ and $\sigma = 1/2$.

1.2 Hessian-free optimization methods

For smooth non-linear function, the following method is generally used: a local method combined with line search work on the scheme $x_{k+1} = x_k + t_k d_k$, where the local method will specify the direction d_k and the line search will specify the step size $t_k \in \mathbb{R}$.

1.2.1 Computing the direction d_k

A desirable property for d_k is that d_k ensures a descent $f(x_{k+1}) < f(x_k)$. Newton methods are such that d_k minimizes a local quadratic approximation of f based on a Taylor expansion, that is $q_f(d) = f(x_k) + g(x_k)^T d + \frac{1}{2} d^T H(x_k) d$ where g denotes the gradient and H denotes the Hessian.

The consists in using the exact solution of local minimization problem $d_k = -H(x_k)^{-1} g(x_k)$.
In practice, other methods are preferred (at least to ensure positive definiteness). The method approximates the Hessian by a matrix H_k as a function of H_{k-1}, x_k, $f(x_k)$ and then d_k solves the system $H_k d = -g(x_k)$. Some implementation may also directly approximate the inverse of the Hessian W_k in order to compute $d_k = -W_k g(x_k)$. Using the Sherman-Morrison-Woodbury formula, we can switch between W_k and H_k.

To determine W_k, first it must verify the secant equation $H_k y_k = s_k$ or $y_k = W_k s_k$ where $y_k = g_{k+1} - g_k$ and $s_k = x_{k+1} - x_k$. To define the $n(n-1)$ terms, we generally impose a symmetry and a minimum distance conditions. We say we have a rank 2 update if $H_k = H_{k-1} + a u u^T + b v v^T$ and a rank 1 update if $H_k = H_{k-1} + a u u^T $. Rank n update is justified by the spectral decomposition theorem.

There are two rank-2 updates which are symmetric and preserve positive definiteness

- DFP minimizes $\min ||H - H_k||_F$ such that $H = H^T$:

$$H_{k+1} = \left(I - \frac{y_k s_k^T}{y_k^T s_k}\right) H_k \left(I - \frac{s_k y_k^T}{y_k^T s_k}\right) + \frac{y_k y_k^T}{y_k^T s_k} \Leftrightarrow W_{k+1} = W_k + \frac{s_k s_k^T}{y_k^T s_k} - \frac{W_k y_k y_k^T W_k^T}{y_k^T W_k y_k}.$$

- BFGS minimizes $\min ||W - W_k||_F$ such that $W = W^T$:

$$\boxed{H_{k+1} = H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \frac{s_k s_k^T}{y_k^T s_k} \Leftrightarrow W_{k+1} = \left(I - \frac{y_k s_k^T}{y_k^T s_k}\right)^T W_k \left(I - \frac{y_k s_k^T}{y_k^T s_k}\right) + \frac{s_k s_k^T}{y_k^T s_k}.}$$

In `R`, the so-called BFGS scheme is implemented in `optim`.

Another possible method (which is initially arised from quadratic problems) is the nonlinear conjugate gradients. This consists in computing directions $(d_0, \dots, d_k)$ that are conjugate with respect to a matrix close to the true Hessian $H(x_k)$. Directions are computed iteratively by $d_k = -g(x_k) + \beta_k d_{k-1}$ for $k > 1$, once initiated by $d_1 = -g(x_1)$. β_k are updated according a scheme:

- $\beta_k = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}}$: Fletcher-Reeves update,
- $\beta_k = \frac{g_k^T (g_k - g_{k-1})}{g_{k-1}^T g_{k-1}}$: Polak-Ribiere update.

There exists also three-term formula for computing direction $d_k = -g(x_k) + \beta_k d_{k-1} + \gamma_k d_t$ for $t < k$. A possible scheme is the Beale-Sorenson update defined as $\beta_k = \frac{g_k^T (g_k - g_{k-1})}{d_{k-1}^T (g_k - g_{k-1})}$ and $\gamma_k = \frac{g_k^T (g_{t+1} - g_t)}{d_t^T (g_{t+1} - g_t)}$ if $k > t+1$ otherwise $\gamma_k = 0$ if $k = t$. See Yuan (2006) for other well-known schemes such as Hestenses-Stiefel, Dixon or Conjugate-Descent. The three updates (Fletcher-Reeves, Polak-Ribiere, Beale-Sorenson) of the (non-linear) conjugate gradient are available in `optim`.

1.2.2 Computing the stepsize t_k

Let $\phi_k(t) = f(x_k + t d_k)$ for a given direction/iterate (d_k, x_k). We need to find conditions to find a satisfactory stepsize t_k. In literature, we consider the descent condition: $\phi_k'(0) < 0$ and the Armijo condition: $\phi_k(t) \leq \phi_k(0) + t c_1 \phi_k'(0)$ ensures a decrease of f. Nocedal & Wright (2006) presents a backtracking (or geometric) approach satisfying the Armijo condition and minimal condition, i.e. Goldstein and Price condition.

- set $t_{k,0}$ e.g. 1, $0 < \alpha < 1$,
- **Repeat** until Armijo satisfied,
  - $t_{k,i+1} = \alpha \times t_{k,i}$.
- **end Repeat**

This backtracking linesearch is available in `optim`.

1.3 Benchmark

To simplify the benchmark of optimization methods, we create a `fitbench` function that computes the desired estimation method for all optimization methods. This function is currently not exported in the package.

```
fitbench <- function(data, distr, method, grad=NULL, control=list(trace=0, REPORT=1, maxit=100
```

2 Numerical illustration with the beta distribution

2.1 Log-likelihood function and its gradient for beta distribution

2.1.1 Theoretical value

The density of the beta distribution is given by

$$f(x; \delta_1, \delta_2) = \frac{x^{\delta_1 - 1}(1-x)^{\delta_2 - 1}}{\beta(\delta_1, \delta_2)},$$

where β denotes the beta function, see the NIST Handbook of mathematical functions https://dlmf.nist.gov/. We recall that $\beta(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. There the log-likelihood for a set of observations $(x_1, \dots, x_n)$ is

$$\log L(\delta_1, \delta_2) = (\delta_1 - 1)\sum_{i=1}^n \log(x_i) + (\delta_2 - 1)\sum_{i=1}^n \log(1 - x_i) + n\log(\beta(\delta_1, \delta_2))$$

The gradient with respect to a and b is

$$\nabla \log L(\delta_1, \delta_2) = \begin{pmatrix} \sum\limits_{i=1}^n \ln(x_i) - n\psi(\delta_1) + n\psi(\delta_1 + \delta_2) \\ \sum\limits_{i=1}^n \ln(1 - x_i) - n\psi(\delta_2) + n\psi(\delta_1 + \delta_2) \end{pmatrix},$$

where $\psi(x) = \Gamma'(x)/\Gamma(x)$ is the digamma function, see the NIST Handbook of mathematical functions https://dlmf.nist.gov/.

2.1.2 `R` implementation

As in the `fitdistrplus` package, we minimize the opposite of the log-likelihood: we implement the opposite of the gradient in `grlnL`. Both the log-likelihood and its gradient are not exported.

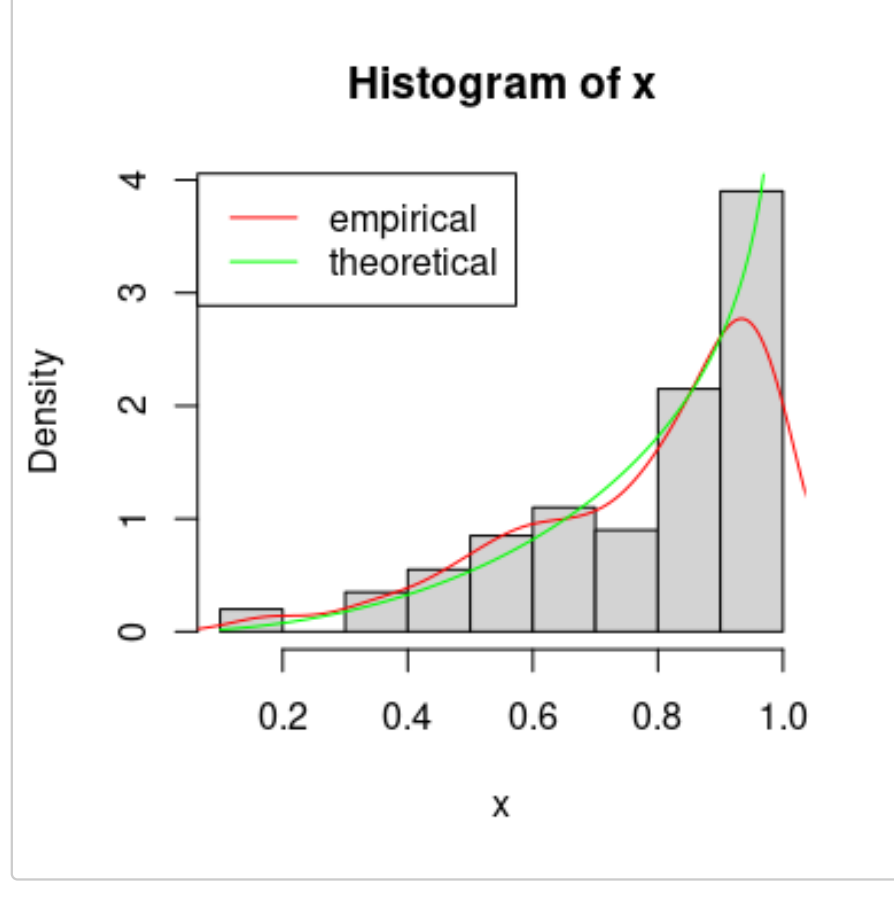
```
lnL <- function(par, fix.arg, obs, ddistnam)
  fitdistrplus:::loglikelihood(par, fix.arg, obs, ddistnam)
grlnlbeta <- fitdistrplus:::grlnlbeta
```

2.2 Random generation of a sample

```
#(1) beta distribution
n <- 200
x <- rbeta(n, 3, 3/4)
grlnlbeta(c(3, 4), x) #test
```

```
## [1] -133  317
```

```
hist(x, prob=TRUE)
lines(density(x), col="red")
curve(dbeta(x, 3, 3/4), col="green", add=TRUE)
legend("topleft", lty=1, col=c("red","green"), leg=c("empirical", "theoretical"))
```



2.3 Fit Beta distribution

Define control parameters.

```
ctr <- list(trace=0, REPORT=1, maxit=1000)
```

Call `mledist` with the default optimization function (`optim` implemented in `stats` package) with and without the gradient for the different optimization methods.

```
unconstropt <- fitbench(x, "beta", "mle", grad=grlnlbeta, lower=0)
```

```
##     BFGS       NM     CGFR     CGPR     CGBS L-BFGS-B     NM-B   G-BFGS 
##       13       13       13       13       13       13       13       13 
##   G-CGFR   G-CGPR   G-CGBS G-BFGS-B   G-NM-B G-CGFR-B G-CGPR-B G-CGBS-B 
##       13       13       13       13       13       13       13       13
```

In the case of constrained optimization, `mledist` permits the direct use of `constrOptim` function (still implemented in `stats` package) that allow linear inequality constraints by using a logarithmic barrier.

Use a exp/log transformation of the shape parameters δ_1 and δ_2 to ensure that the shape parameters are strictly positive.

```
dbeta2 <- function(x, shape1, shape2, log)
  dbeta(x, exp(shape1), exp(shape2), log=log)
#take the log of the starting values
startarg <- lapply(fitdistrplus:::start.arg.default(x, "beta"), log)
#redefine the gradient for the new parametrization
grbetaexp <- function(par, obs, ...)
    grlnlbeta(exp(par), obs) * exp(par)


expopt <- fitbench(x, distr="beta2", method="mle", grad=grbetaexp, start=startarg)
```

```
##   BFGS     NM   CGFR   CGPR   CGBS G-BFGS G-CGFR G-CGPR G-CGBS 
##     13     13     13     13     13     13     13     13     13
```

```
#get back to original parametrization
expopt[c("fitted shape1", "fitted shape2"), ] <- exp(expopt[c("fitted shape1", "fitted shape2")
```

Then we extract the values of the fitted parameters, the value of the corresponding log-likelihood and the number of counts to the function to minimize and its gradient (whether it is the theoretical gradient or the numerically approximated one).

2.4 Results of the numerical investigation

Results are displayed in the following tables: (1) the original parametrization without specifying the gradient (`-B` stands for bounded version), (2) the original parametrization with the (true) gradient (`-B` stands for bounded version and `-G` for gradient), (3) the log-transformed parametrization without specifying the gradient, (4) the log-transformed parametrization with the (true) gradient (`-G` stands for gradient).

	BFGS	NM	CGFR	CGPR	CGBS	L-BFGS-B	NM-B
fitted shape1	2.665	2.664	2.665	2.665	2.665	2.665	2.665
fitted shape2	0.731	0.731	0.731	0.731	0.731	0.731	0.731
fitted loglik	114.165	114.165	114.165	114.165	114.165	114.165	114.165
func. eval. nb.	23.000	47.000	211.000	263.000	183.000	11.000	47.000
grad. eval. nb.	5.000	NA	53.000	69.000	47.000	11.000	NA
time (sec)	0.005	0.005	0.031	0.037	0.027	0.007	0.009

	G-BFGS	G-CGFR	G-CGPR	G-CGBS	G-BFGS-B	G-NM-B	G-CGFR-B	G-CGPR-B	G-CGBS-B
fitted shape1	2.665	2.665	2.665	2.665	2.665	2.665	2.665	2.665	2.665
fitted shape2	0.731	0.731	0.731	0.731	0.731	0.731	0.731	0.731	0.731
fitted loglik	114.165	114.165	114.165	114.165	114.165	114.165	114.165	114.165	114.165
func. eval. nb.	20.000	249.000	225.000	138.000	25.000	47.000	263.000	188.000	176.000
grad. eval. nb.	5.000	71.000	69.000	43.000	5.000	NA	69.000	59.000	47.000
time (sec)	0.012	0.081	0.078	0.052	0.018	0.016	0.092	0.082	0.072

	BFGS	NM	CGFR	CGPR	CGBS
fitted shape1	2.665	2.664	2.665	2.665	2.665
fitted shape2	0.731	0.731	0.731	0.731	0.731
fitted loglik	114.165	114.165	114.165	114.165	114.165
func. eval. nb.	18.000	41.000	131.000	116.000	134.000
grad. eval. nb.	5.000	NA	27.000	29.000	35.000
time (sec)	0.007	0.005	0.019	0.018	0.021

	G-BFGS	G-CGFR	G-CGPR	G-CGBS
fitted shape1	2.665	2.665	2.665	2.665
fitted shape2	0.731	0.731	0.731	0.731
fitted loglik	114.165	114.165	114.165	114.165
func. eval. nb.	20.000	175.000	125.000	112.000
grad. eval. nb.	5.000	39.000	41.000	35.000
time (sec)	0.014	0.052	0.049	0.045

Using `llsurface`, we plot the log-likehood surface around the true value (green) and the fitted parameters (red).

```
llsurface(min.arg=c(0.1, 0.1), max.arg=c(7, 3),
          plot.arg=c("shape1", "shape2"), nlev=25,
          plot.np=50, data=x, distr="beta", back.col = FALSE)
```

```
## Warning in plot.window(xlim, ylim, ...): "plot.np" n'est pas un paramètre
## graphique
```

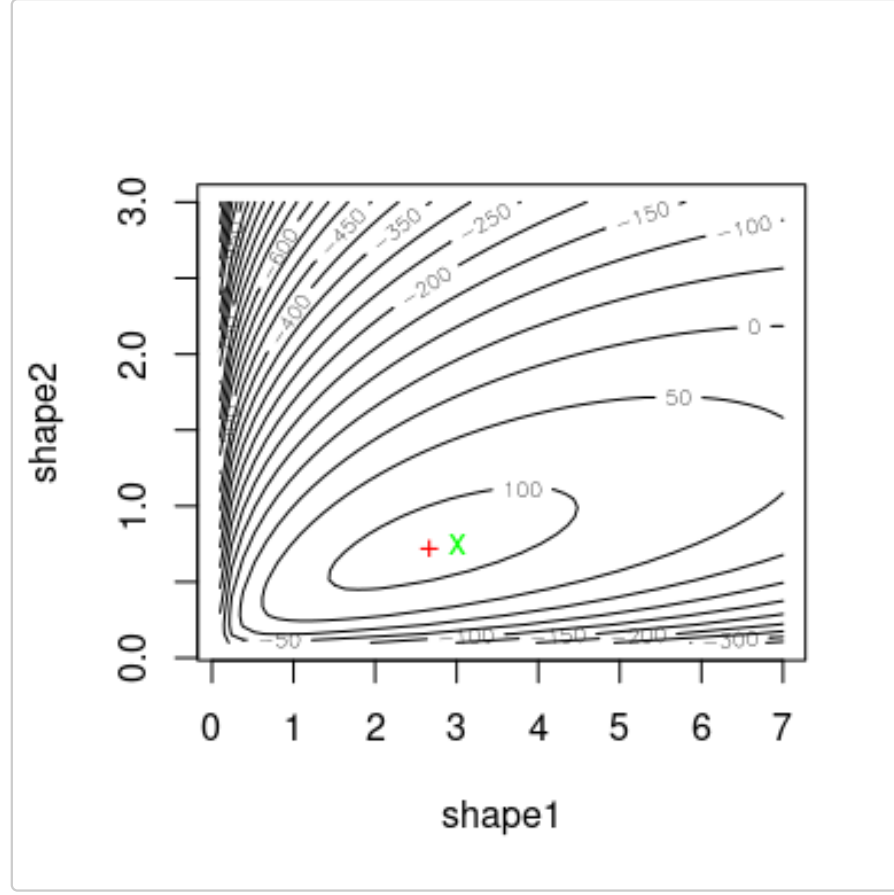
```
## Warning in title(...): "plot.np" n'est pas un paramètre graphique
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "plot.np" n'est pas
## un paramètre graphique

## Warning in axis(side = side, at = at, labels = labels, ...): "plot.np" n'est pas
## un paramètre graphique
```

```
## Warning in box(...): "plot.np" n'est pas un paramètre graphique
```

```
points(unconstropt[1,"BFGS"], unconstropt[2,"BFGS"], pch="+", col="red")
points(3, 3/4, pch="x", col="green")
```

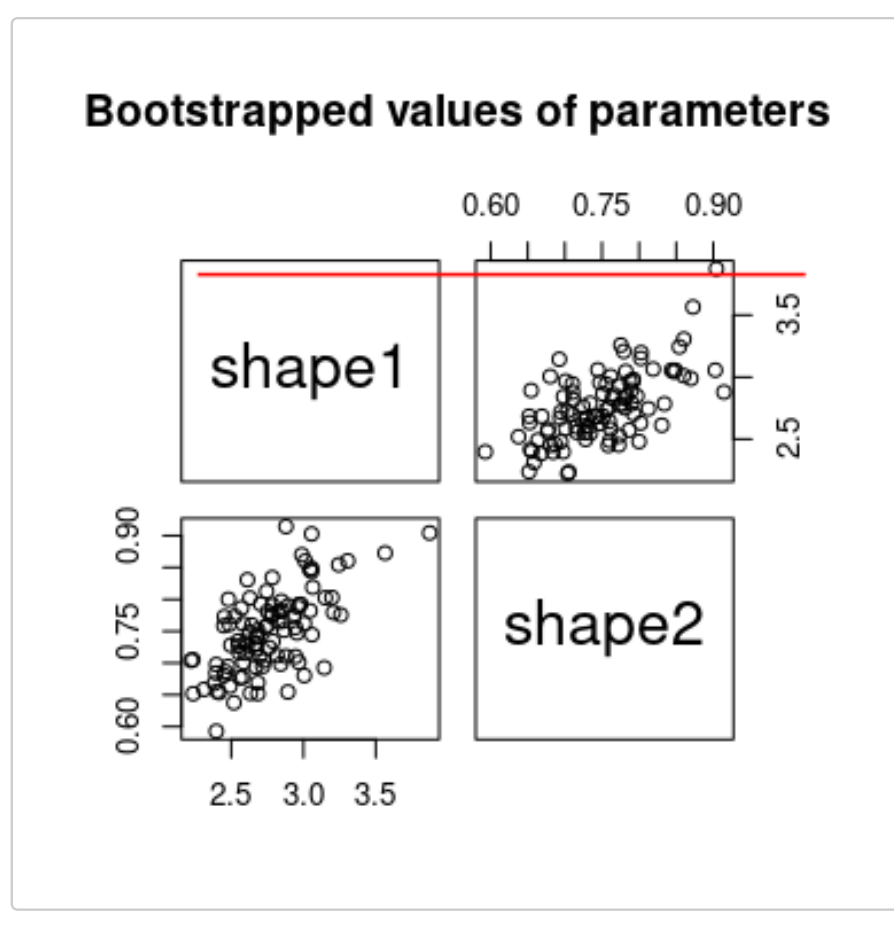


We can simulate bootstrap replicates using the `bootdist` function.

```
b1 <- bootdist(fitdist(x, "beta", method="mle", optim.method="BFGS"), niter=100, parallel="sn
summary(b1)
```

```
## Parametric bootstrap medians and 95% percentile CI 
##          Median  2.5% 97.5%
## shape1   2.73 2.272 3.283
## shape2   0.75 0.652 0.888
```

```
plot(b1)
abline(v=3, h=3/4, col="red", lwd=1.5)
```



3 Numerical illustration with the negative binomial distribution

3.1 Log-likelihood function and its gradient for negative binomial distribution

3.1.1 Theoretical value

The p.m.f. of the Negative binomial distribution is given by

$$f(x; m, p) = \frac{\Gamma(x+m)}{\Gamma(m)x!} p^m (1-p)^x,$$

where Γ denotes the beta function, see the NIST Handbook of mathematical functions https://dlmf.nist.gov/. There exists an alternative representation where $\mu = m(1-p)/p$ or equivalently $p = m/(m+\mu)$. Thus, the log-likelihood for a set of observations $(x_1, \dots, x_n)$ is

$$\log L(m, p) = \sum_{i=1}^n \log\Gamma(x_i + m) - n\log\Gamma(m) - \sum_{i=1}^n \log(x_i!) + mn\log(p) + \sum_{i=1}^n x_i \log(1-p)$$

The gradient with respect to m and p is

$$\nabla \log L(m, p) = \begin{pmatrix} \sum_{i=1}^n \psi(x_i + m) - n\psi(m) + n\log(p) \\ mn/p - \sum_{i=1}^n x_i/(1-p) \end{pmatrix},$$

where $\psi(x) = \Gamma'(x)/\Gamma(x)$ is the digamma function, see the NIST Handbook of mathematical functions https://dlmf.nist.gov/.

3.1.2 `R` implementation

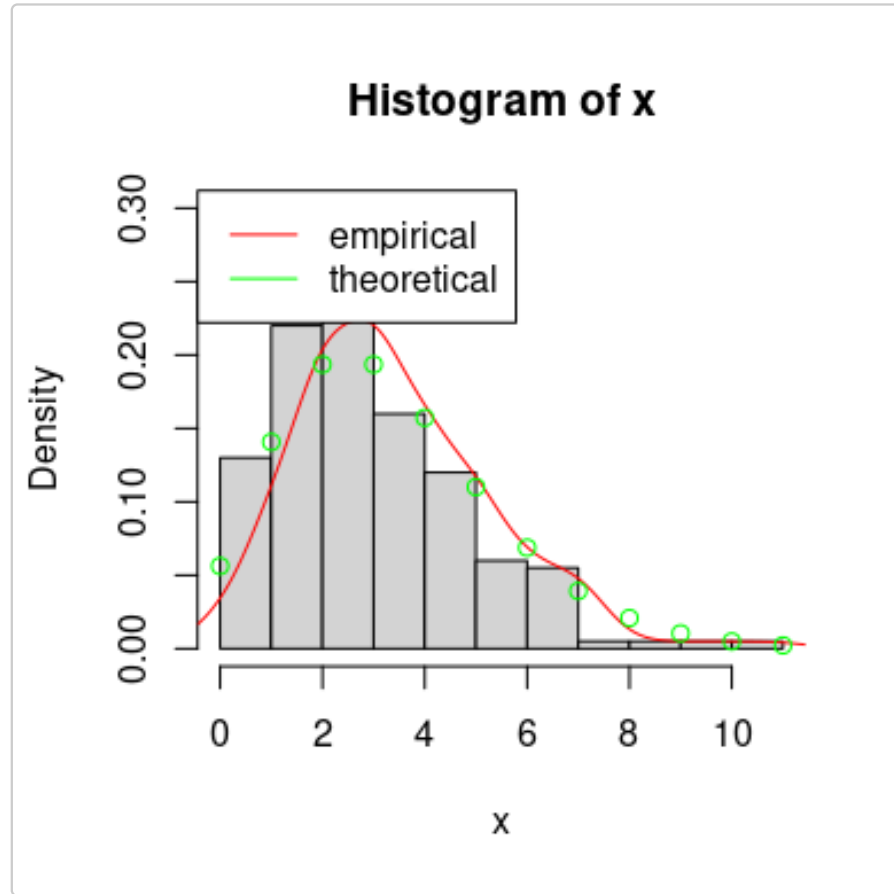
As in the `fitdistrplus` package, we minimize the opposite of the log-likelihood: we implement the opposite of the gradient in `grlnL`.

```
grlnlNB <- function(x, obs, ...)
{
  m <- x[1]
  p <- x[2]
  n <- length(obs)
  c(sum(psigamma(obs+m)) - n*psigamma(m) + n*log(p),
    m*n/p - sum(obs)/(1-p))
}
```

3.2 Random generation of a sample

```
#(1) beta distribution
n <- 200
trueval <- c("size"=10, "prob"=3/4, "mu"=10/3)
x <- rnbinom(n, trueval["size"], trueval["prob"])

hist(x, prob=TRUE, ylim=c(0, .3))
lines(density(x), col="red")
points(min(x):max(x), dnbinom(min(x):max(x), trueval["size"], trueval["prob"]), col="green")
legend("topleft", lty=1, col=c("red","green"), leg=c("empirical", "theoretical"))
```



3.3 Fit a negative binomial distribution

Define control parameters and make the benchmark.

```
ctr <- list(trace=0, REPORT=1, maxit=1000)
unconstropt <- fitbench(x, "nbinom", "mle", grad=grlnlNB, lower=0)
```

```
##     BFGS       NM     CGFR     CGPR     CGBS L-BFGS-B     NM-B   G-BFGS 
##       13       13       13       13       13       13       13       13 
##   G-CGFR   G-CGPR   G-CGBS G-BFGS-B   G-NM-B G-CGFR-B G-CGPR-B G-CGBS-B 
##       13       13       13       13       13       13       13       13
```

```
unconstropt <- rbind(unconstropt, "fitted prob"=unconstropt["fitted mu",] / (1+unconstropt["fit
```

In the case of constrained optimization, `mledist` permits the direct use of `constrOptim` function (still implemented in `stats` package) that allow linear inequality constraints by using a logarithmic barrier.

Use a exp/log transformation of the shape parameters δ_1 and δ_2 to ensure that the shape parameters are strictly positive.

```
dnbinom2 <- function(x, size, prob, log)
  dnbinom(x, exp(size), 1/(1+exp(-prob)), log=log)
#transform starting values
startarg <- fitdistrplus:::start.arg.default(x, "nbinom")
startarg$mu <- startarg$size / (startarg$size+startarg$mu)
startarg <- list(size=log(startarg[[1]]), prob=log(startarg[[2]]/(1-startarg[[2]])))

#redefine the gradient for the new parametrization
Trans <- function(x)
  c(exp(x[1]), plogis(x[2]))
grNBexp <- function(par, obs, ...)
    grlnlNB(Trans(par), obs) * c(exp(par[1]), plogis(x[2])*(1-plogis(x[2])))

expopt <- fitbench(x, distr="nbinom2", method="mle", grad=grNBexp, start=startarg)
```

```
##   BFGS     NM   CGFR   CGPR   CGBS G-BFGS G-CGFR G-CGPR G-CGBS 
##     13     13     13     13     13     13     13     13     13
```

```
#get back to original parametrization
expopt[c("fitted size", "fitted prob"), ] <- apply(expopt[c("fitted size", "fitted prob"), ], 2
```

Then we extract the values of the fitted parameters, the value of the corresponding log-likelihood and the number of counts to the function to minimize and its gradient (whether it is the theoretical gradient or the numerically approximated one).

3.4 Results of the numerical investigation

Results are displayed in the following tables: (1) the original parametrization without specifying the gradient (`-B` stands for bounded version), (2) the original parametrization with the (true) gradient (`-B` stands for bounded version and `-G` for gradient), (3) the log-transformed parametrization without specifying the gradient, (4) the log-transformed parametrization with the (true) gradient (`-G` stands for gradient).

	BFGS	NM	CGFR	CGPR	CGBS	L-BFGS-B	NM-B
fitted size	61.944	68.138	61.969	62.475	62.878	61.944	67.397
fitted mu	3.425	3.425	3.425	3.425	3.425	3.425	3.425
fitted loglik	-401.612	-401.611	-401.612	-401.612	-401.612	-401.612	-401.611
func. eval. nb.	2.000	37.000	2999.000	2663.000	2610.000	2.000	0.000
grad. eval. nb.	1.000	NA	1001.000	1001.000	1001.000	2.000	NA
time (sec)	0.002	0.003	0.244	0.248	0.248	0.002	0.003
fitted prob	0.774	0.774	0.774	0.774	0.774	0.774	0.774

	G-BFGS	G-CGFR	G-CGPR	G-CGBS	G-BFGS-B	G-NM-B	G-CGFR-B	G-CGPR-B	G-CGBS-B
fitted size	61.944	61.944	61.944	61.944	61.944	67.397	61.944	61.944	61.944
fitted mu	3.425	3.425	3.425	3.425	3.425	3.425	3.425	3.425	3.425
fitted loglik	-401.612	-401.612	-401.612	-401.612	-401.612	-401.611	-401.612	-401.612	-401.612
func. eval. nb.	26.000	233.000	174.000	233.000	0.000	0.000	0.000	0.000	0.000
grad. eval. nb.	1.000	15.000	11.000	15.000	NA	NA	NA	NA	NA
time (sec)	0.011	0.014	0.010	0.015	0.002	0.003	0.013	0.010	0.013
fitted prob	0.774	0.774	0.774	0.774	0.774	0.774	0.774	0.774	0.774

	BFGS	NM	CGFR	CGPR	CGBS
fitted size	61.946	67.787	63.450	67.940	67.885
fitted prob	0.948	0.952	0.949	0.952	0.952
fitted loglik	-401.612	-401.611	-401.612	-401.611	-401.611
func. eval. nb.	6.000	47.000	4001.000	3728.000	327.000
grad. eval. nb.	1.000	NA	1001.000	1001.000	87.000
time (sec)	0.006	0.004	0.302	0.235	0.027

	G-BFGS	G-CGFR	G-CGPR	G-CGBS
fitted size	61.944	61.944	61.944	61.944
fitted prob	0.948	0.948	0.948	0.948
fitted loglik	-401.612	-401.612	-401.612	-401.612
func. eval. nb.	21.000	43.000	42.000	42.000
grad. eval. nb.	1.000	3.000	3.000	3.000
time (sec)	0.009	0.003	0.003	0.003

Using `llsurface`, we plot the log-likehood surface around the true value (green) and the fitted parameters (red).

```
llsurface(min.arg=c(5, 0.3), max.arg=c(15, 1),
          plot.arg=c("size", "prob"), nlev=25,
          plot.np=50, data=x, distr="nbinom", back.col = FALSE)
```

```
## Warning in plot.window(xlim, ylim, ...): "plot.np" n'est pas un paramètre
## graphique
```

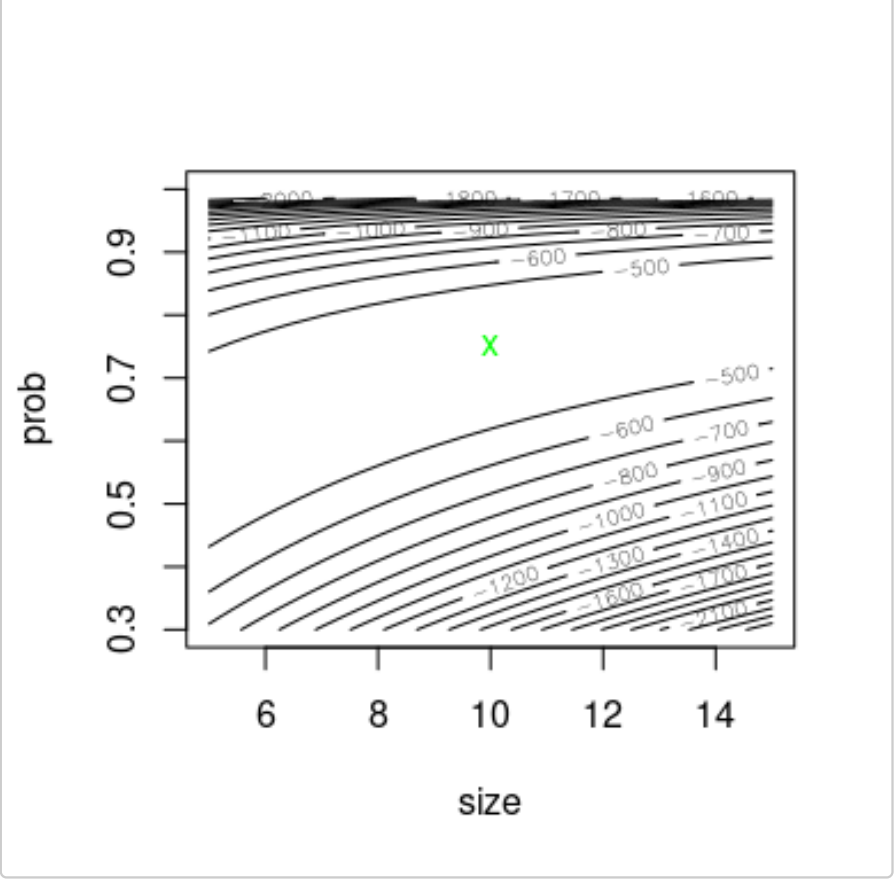
```
## Warning in title(...): "plot.np" n'est pas un paramètre graphique
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "plot.np" n'est pas
## un paramètre graphique

## Warning in axis(side = side, at = at, labels = labels, ...): "plot.np" n'est pas
## un paramètre graphique
```

```
## Warning in box(...): "plot.np" n'est pas un paramètre graphique
```

```
points(unconstropt["fitted size","BFGS"], unconstropt["fitted prob","BFGS"], pch="+", col="red"
points(trueval["size"], trueval["prob"], pch="x", col="green")
```

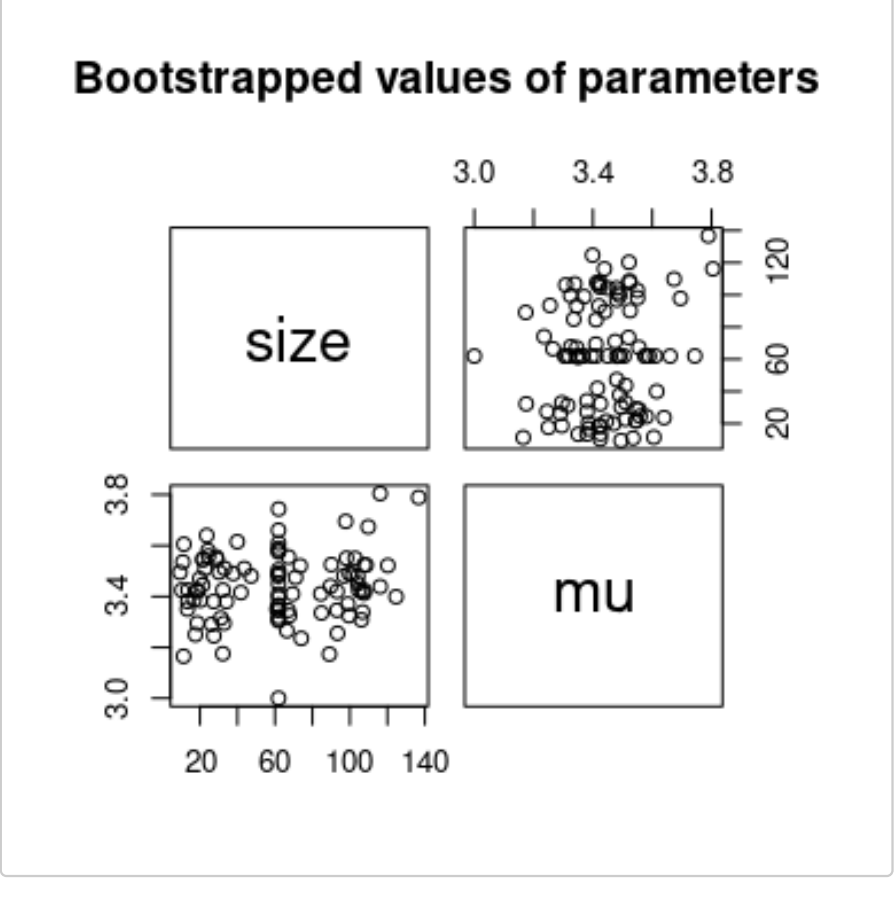


We can simulate bootstrap replicates using the `bootdist` function.

```
b1 <- bootdist(fitdist(x, "nbinom", method="mle", optim.method="BFGS"), niter=100, parallel="s
summary(b1)
```

```
## Parametric bootstrap medians and 95% percentile CI 
##      Median  2.5%  97.5%
## size  61.95 11.05 118.32
## mu     3.43  3.17   3.72
## 
## The estimation method converged only for 76 among 100 iterations
```

```
plot(b1)
abline(v=trueval["size"], h=trueval["mu"], col="red", lwd=1.5)
```



4 Conclusion

Based on the two previous examples, we observe that all methods converge to the same point. This is rassuring.
However, the number of function evaluations (and the gradient evaluations) is very different from a method to another. Furthermore, specifying the true gradient of the log-likelihood does not help at all the fitting procedure and generally slows down the convergence. Generally, the best method is the standard BFGS method or the BFGS method with the exponential transformation of the parameters. Since the exponential function is differentiable, the asymptotic properties are still preserved (by the Delta method) but for finite-sample this may produce a small bias.